

Chatbot-based Interview Simulator: A Feasible Approach to Train Novice Requirements Engineers

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Abstract—Introduction: Although the interview is the most important and widely used requirements elicitation technique, novice engineers do not receive adequate training in Requirements Engineering (RE) courses.

Objectives: Develop an AI-based interview simulator for helping novice requirements engineers in gaining interview skills.

Methods: The research is based on the Design Science Methodology for Information Systems. The simulator is the outcome of six cycles; in each cycle, a proof of concept with additional features is created. Each cycle finishes with evaluation and improvement suggestions.

Results: The simulator has been tested with students and results have been promising. The interview simulator understands context-free questions, retrieving the right information related to RE concepts such as goals, tasks, users, benefits, and constraints. The simulator also answers questions based on the context, makes summaries of the conversation, responds to meta-questions, and adds ambiguity and incompleteness to the conversation.

Conclusions: The results have been promising. The simulator has been tested with degree and master level students. They were able to create a requirements specification using the simulator, and the feedback has been generally positive. The simulator will be tested in a real RE course in the academic year 2020-2021. Once it proves effective in the classroom, it will be opened to the RE community for free use and improvement.

Index Terms—Requirements elicitation, chatbots, interviews, training.

I. INTRODUCTION

Requirements elicitation (REI) is a crucial phase of requirements engineering [1], which strongly influences the quality of final product [2]. Communication skills are fundamental to elicitation; hence people with strong communication skills are more effective in the requirements elicitation [3]. Users and customers should have enough skills to express their needs to the requirements engineers, and obviously, requirements engineers should be specialists in acquiring the necessary domain-specific knowledge.

REI comprises diverse activities and multiple techniques to perform these activities. There exist many requirements elicitation techniques such as *task analysis*, *questionnaires*, *domain analysis*, *interviews*, *observation*, etc. Elicitation techniques are selected depending on system type, purpose, time, budget, etc. Interviews are the most widely used elicitation technique for requirements acquisition [4], and they are considered as one of the most effective elicitation techniques [5].

Interviews look easy to conduct because they are, in essence, a one-to-one conversation. However, an *effective* interview is not a free conversation. On the one hand, interviews require the realization of several tasks, e.g., scheduling, script preparation, and consolidation. On the other hand, their success significantly depends on the skill of the interviewer to know what are right questions and when to ask them, and from whom to ask [4]. Interviews are a soft skill but it is not acquired for free.

It is difficult for REI students to develop interview skills during their courses. The problems are multi-fold: (1) students usually have limited domain knowledge, (2) interview practice is time-consuming, and (3) large groups prevent instructors to conduct even a single interview with each student. At least in our experience, interview training in academia restricts to theory or short-time exercises.

Similar problems have been reported in other sciences. Fitzmaurice et al. [6] highlight that young doctors have little chances for interviewing patients to acquire interview skills. Yang and Evans [9] discuss the need for training for the use of a newly introduced educational application. Kowalski, Pavlovskaya, and Goldstein [10] explains the training of security personnel for effective communication.

An approach that has been explored by several researchers to improve the engineers' interviewing skills is to develop interview simulators, likewise e.g., airline pilots are trained in flight simulators. However, we are not aware that any of these simulators are currently available. It is even possible that they did not proceed beyond the research stage, as the natural language processing technology has developed at a rather slow pace in the past years. The situation has changed with the rise of computing power and the expansion of AI-enabled technologies, e.g., digital assistants such as chatbots are commonplace [7] for customer assistance in many web pages and traditional phone systems.

In this paper, we aim to develop an AI-based interview simulator for helping novice requirements engineers to acquire interview skills. More specifically, our research goals are:

- 1) Create a chatbot that engages in a natural conversation with requirements engineers.
- 2) Emphasize good interview practice during the conversation, e.g., using context-free questions [8], [9], making

summaries [10], etc.

- 3) Introduce natural language pitfalls, e.g., ambiguity, incompleteness, etc. in the engineer-chatbot conversation.
- 4) Elaborate on a design theory for chatbots applied to REI techniques.

This research applies the *Design Science* [11] methodology, due to its requirement of developing a working software product. We have carried out six *Design Science* cycles and the results are promising. The interview simulator understands context-free, context-related, and meta-questions. It also randomly introduces ambiguity and incompleteness in the conversation. We have also created the first draft of a generic design for REI tools using IBM Watson technologies.

The remainder of this paper is organized as follows: In section II, we provide relevant background information. In section III, we introduce the *Design Science* methodology, that we use for the design and implementation of the simulator. Section IV summarizes the *Design Science* cycles and the simulator architecture. In section V, we report the evaluation of the simulator. Finally, in sections VI and VII, we report the threats to validity, conclusions and future works, respectively.

II. BACKGROUND

In section II-A, we review how different sciences have addressed the need that practitioners receive adequate interview training. In section II-B, we discuss the interview training process in RE. In the remaining sections, we discuss chatbot-related issues, i.e., the educational value of chatbots (section II-C), and the IBM's Watson technologies (section II-D).

A. Interview training

The need for adequate interview training has been raised in several disciplines, such as education, security and health.

Yang and Evans [12] discuss the need for training for the use of a newly introduced educational application. A university has introduced a system for managing library resources that helps teachers to generate reading lists and students to access those lists. The problem is that a significant number of users including students, teachers, and library staff do not know how to utilize service properly. They developed a chatbot that acts as a virtual tutor to instruct and train users to solve basic issues.

Kowalski, Pavlovska, and Goldstein [13] explain the training of security personnel in a company. Such training was mostly based on intranet e-learning, but traditional web-based learning processes have not been proved effective. This study used chatbot technologies to enhance security awareness training. The results of case studies conducted on the company suggest that the attitudes of the respondents appear to be more positive to security when chatbots are used instead of the current traditional e-learning security training courses.

Fitzmaurice et al. [6] highlights that young doctors do not have enough chances for interviewing patients to acquire interview skills. In general, students are supposed to interview many patients with acute and severe psychiatric disorders. Unfortunately, patients are reluctant to be interviewed by

multiple students. To solve this problem, an online interactive e-learning simulator called 'Virtual Interviews for Students Interacting Online for Psychiatry (VISION)' has been developed. Students can use simulator anytime to learn about their profession and apply their learning outcomes in real life.

B. Interview training in RE

The problems associated to interview training suggest the need for some computerized support to the education of novice requirements engineers. Computer-Aided Software Engineering (CASE) tools have been applied to a diversity of software engineering in a variety of ways. Only a few tools have been specifically created for REI [10]

Debnath and Spoletini [14] propose an intent-based multimodal virtual conversational agent to eliminate the issues of scheduling interviews for the students during the class. Students can use the virtual customer in their free time; the simulator also generates transcripts that students can utilize to review their performance. An initial exploratory study with undergraduate students has been performed for the evaluation of the first prototype. As a result, they report that participants who used the system before conducting interviews with customers performed well in comparison to those who did not use the system.

Rietz and Maedche [15] present a self-elicitation system that uses the laddering technique to guide novice requirements engineers during the elicitation process. The system employs randomized rephrasing of questions based on grounded guidelines for interviews. An experimental study has been proposed by authors to evaluate the system.

Gilvaz & do Prado Leite [16] presents an interview assistant to help software engineers during the interview process. The assistant uses Wetherbe's indirect questioning approach [17] for interviewing and provides automated support for the software engineers in eliciting necessary information for the information systems.

A business interview simulator has been developed by Bollweg, Shahriar, Stemmermann & Weber [18] as part of their Enterprise resource planning MOOC course. Their primary purpose in developing a simulator is to provide students with interviewing skills so that they can efficiently conduct interviews (face-to-face or online) with managers from real companies later in the technical phase of course. The simulator has been tested on 38 students enrolled in the course. Students were asked to fill questionnaires right after using the simulator. The results indicate that the simulator performed well technically to help students gaining interview skills as compared to traditional e-learning process.

Yamanaka and Komiya [19] illustrates a method to navigate interview-driven software requirements elicitation. The method uses the progress management table to keep users on track. To check the effectiveness of the proposed method an experiment was conducted on SE student with and without navigation method. As a result, they clarify that the user requirements elicitation work depends on the business knowledge and work experience of each individual.

Kato et al. [10] presents a model of navigating interview processes for eliciting requirements to help novice analysts. Their model stores expert analyst's data in the forms of thesaurus, state transition model, and template, then they use that information for the interview navigation process. The navigation process help analysts to specify SRSs in IEEE 830-1993 format. They validated their model by comparing the SRSs created by both novice and expert analyst and results confirm that the navigation method will help both novice and expert analysts as well. However, the development of supporting system based on the method is yet to be done.

Stanica et al. [20] developed a Virtual Reality (VR) Job Interview Simulator to help software engineers increase their job interview performances by practicing their hard (skills related to job) and soft (social competencies, communication skills, and personality-related traits) skills. The interview simulation process is performed using VR based environments. However, the product is still under development soon it will be validated with real candidates.

C. The educational value of chatbots

Chatbots have been widely used in different fields but their use in academia is limited despite their potential to support many educational activities. Yang and Evans [12] discuss three chatbots with the purpose: (1) to support the delivery of a Master's course simulation game, (2) to support the training and use of a newly introduced educational application, and (3) to improve the processing of help-desk requests within a university department. The results indicate that AI-based chatbots are quite useful in the field of education but still there are some challenges in the creation of chatbots i.e. chatbot development platforms are not originally designed to support educational activities.

D. IBM Watson technologies

We have chosen IBM Watson technologies to develop our interview simulator due to its capabilities to adopt AI techniques on different domains with high accuracy [21]. Our interview simulator uses specifically the products: (1) Watson Assistant, (2) Watson Discovery Service, (3) Watson Knowledge Studio, and (4) IBM cloud Functions. We describe them briefly below.

1) *Watson Assistant*: is IBM's artificial intelligence-based product that lets developers build, train, and deploy smart assistants for businesses. The assistants can be deployed standalone or part of any application. Unlike traditional chatbots that frustrate humans when complexity increases, the Watson Assistant is rather smart and provide external search support [22].

2) *Watson Discovery*: is an intelligence-based search technology with machine learning and natural language processing capabilities that work accurately to generate meaningful results when applied to huge enterprise data. Moreover, it also provides support to easily create cognitive, cloud-based exploration applications that uncover actionable insights embedded in unstructured data - including own proprietary data, as well as public and third-party data [23].

3) *Watson Knowledge Studio*: is a cloud-based application that lets developers and domain experts work together to create and train custom machine learning models that could be used to recognize concepts and relationships in unstructured text. Moreover, it also provides support to use the models in IBM Watson Discovery, IBM Watson Natural Language Understanding, and IBM Watson Explorer [24].

4) *Cloud Functions*: are Functions-as-a-Service platform based on Apache OpenWhisk, that runs application code without servers [25].

III. METHODOLOGY

AI-enabled technologies, particularly chatbots, have been used for interview support and training. Some efforts [10], [16], [18], [19] have been made in RE to provide support to requirements engineering during REI, but very little in what regards novice engineers' training and interviewing skills acquisition.

We aim to develop an interview simulator, in line with the works of [18], [6], [20]. As this research's goal is to design a new artifact, we will apply the *Design Science* methodology [11].

Design Science was created in the Information Systems (IS) discipline. IS involve people, organizations, and technology. Developing IS requires proper analysis of all connected entities. Hevner [11] argues that this discipline is characterized by two separate paradigms: behavior science and design science. The first one seeks the creation or justification of theories to explain or predict human behavior. The second one is the engineering approach focused and aims to build inventions that solve people's and organization's problems effectively and efficiently.

While the routine design is the application of existing knowledge to organizational problems, *Design Science* involves finding new solutions to previously unsolved problems or better and more efficient solutions to previously solved problems [26]. Our research matches one of these two situations, depending on each one's interpretation of the current state of the art.

Fig. 1 shows the design science framework for this research. The interview simulator is developed using an iterative approach, creating a sequence of proof-of-concept artifacts. Artifacts are refined through the feedback received from different evaluation mechanisms. Once evaluated, the information flows back both to the environment (reporting about e.g., feasibility, scope limitations, etc.) and to the knowledge base (increasing our knowledge about the research area).

IV. RESULTS

The simulator consists of six cycles. In this section, we will report the work carried out during these cycles, followed by the chatbot architecture and evaluation.

A. Design Science process

1) *First research cycle*: It was focused on designing a basic chatbot with the following features:

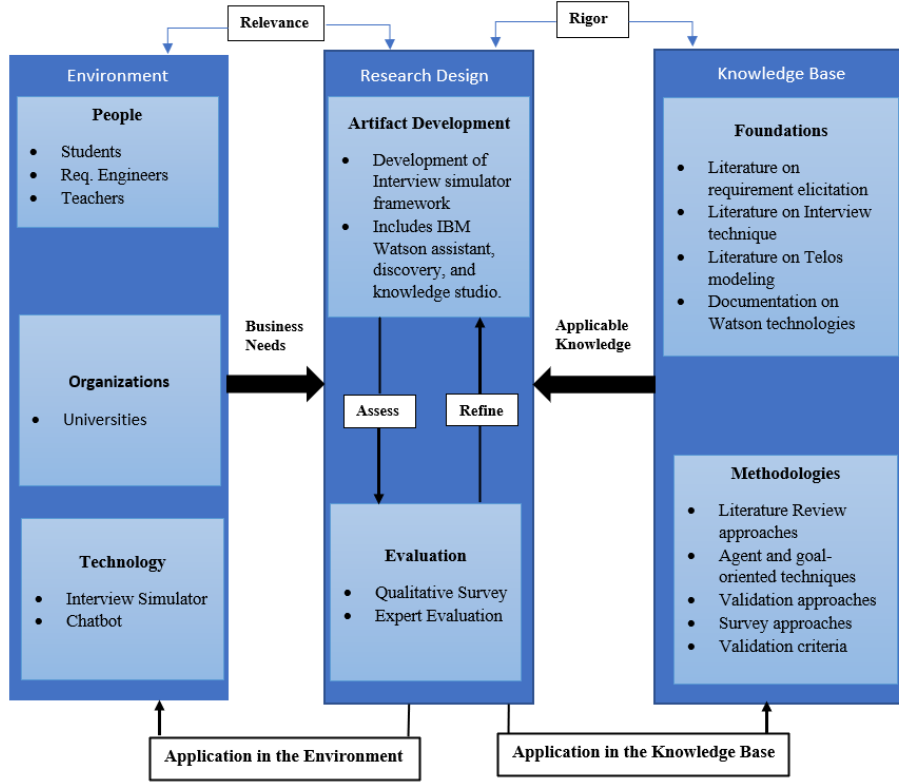


Fig. 1. Research design science framework (based on [11]).

- Accept as input one IEEE-Std-830 [27] requirements specification. We have chosen a short specification available at the nlrpBENCH¹: The Display Management System².
- Answer queries related to **goals, tasks, and users** of the system described in the requirements specification.

This second point deserves clarification. The design of the interview simulator relies on a RE concept model. There are several RE model proposals, e.g., Sutcliffe and Maiden [28, Fig. 9], Diamantopoulos et al., [29, Fig. 2], the FRISCO Report [30], etc. However, they are too complex for this research.

We defined a basic RE concept model during the first research cycle, which will be extended in later research cycles. This model contains three concepts: goals, tasks and users, along with two relationships (tasks operationalize goals and users perform task). This RE concept model can be seen as an *L1*-model based on a *L2*-model (meta-model) with just two components: *classes* and *relationships*. This model can be easily implemented in many tools (such as the Knowledge Studio, see next cycle), and formalized using a variety of languages, e.g., TELOS [31].

The features were implemented using the Watson Assistant, Watson Discovery, and IBM Cloud functions. Watson Assis-

tant provides support to identify the interviewee’s intents and the cloud function runs queries on the discovery service. The discovery service returns the information related to the user intent, due to its ability to assign meaning to text documents.

2) *Second research cycle*: This cycle aimed at making the assistant more robust; thus, we did not introduce new features but improve the current ones. The main lines of action were the introduction of Watson Knowledge Studio and its machine learning model. The model was deployed on Watson Discovery. The Assistant and the Cloud functions were updated to accommodate the changes.

As described below in Section IV-B, Knowledge Studio leverages the creation of conceptual models using machine-learning (ML) techniques. We have created a conceptual model (a *L0*-model, that is, an instance of the *L1*-RE concept model) that contains domain-specific concepts (related to *Display Management*). This conceptual model has been created using a large set of domain-specific documents (our training dataset is available at [32, Annex K]).

Watson Knowledge Studio is a key component of the interview simulator. On the one side, Watson Discovery works on a single domain-specific document (the target SRS). The conceptual model improves Watson Discovery’s abilities to return the correct information. On the other, the conceptual models enable the simulator to work in different problem domains.

¹http://nlrp.ipd.kit.edu/index.php/Main_Page

²http://nlrp.ipd.kit.edu/index.php/Display_Management_System

3) *Third research cycle*: The third cycle is focused on adding more intents and improving the performance of the machine learning model with the use of a bigger training dataset. To speed up the annotation process for model training and reduce the human annotator workload, we created a domain-specific dictionary for each entity type (goals, tasks, etc.).

4) *Fourth research cycle*: In the fourth cycle, we focused on making changes in Watson Discovery and cloud code to respond in chunks for relevant intents rather than answering in a single paragraph. We pick the most relevant response of the query and return it to the assistant.

5) *Fifth research cycle*: This cycle was dedicated to adding features to improve the conversation flow, allowing the interview simulator to engage in a more natural conversation. We introduced support to answer engineers' meta-questions, such as "is there anything else" or "Am I correct". We keep a record of the interaction with the simulator for making summaries, and we use Discovery language queries for answering meta-questions.

6) *Sixth research cycle*: In this last iteration, we focused on introducing natural language pitfalls in the simulator, i.e., ambiguity and incompleteness. For adding ambiguity we remove entities randomly from the response; for adding incompleteness, we remove sentences randomly.

B. Simulator architecture

Fig. 2 depicts the interview simulator architecture (details are available in [32]). The user initiates the conversation and interacts with the simulator using the provided web interface (arrow labeled as "2"). The Watson Assistant collects the inputs (arrow "3") and identifies the intent. For instance, when the user enters the question:

What are the system goals?

the Watson Assistant recognizes the question due to examples provided previously, and identifies that the requirements engineer wants to find out about system #goals.

When an intent is identified, the corresponding information is retrieved. However, the information about the RE concepts is not readily available to the simulator yet. As indicated before, such information is represented as a regular IEEE-Std-830 requirements specification ("target SRS"), usually in PDF format. Using a document written in natural language is highly convenient for instructors and trainees. In turn, the correct information should be extracted from such a document. This is possible thanks to the Watson Discovery service's document understanding technology.

Information extraction is triggered by the Watson Assistant using IBM Cloud functions (arrows "4" and "5"). In the case of the example question given before, the answer provided by the simulator is:

The goal of this project is developing a system capable to manage the displayed contents over the Internet, so that

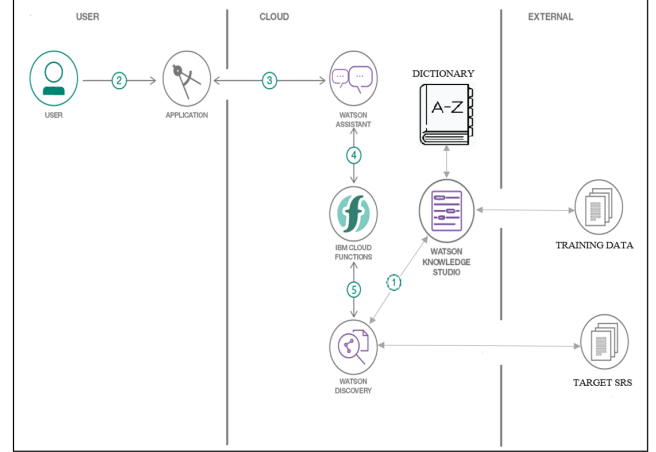


Fig. 2. Chatbot Architecture (following [33]).

an administrator may make changes from anywhere.

Watson Discovery's ability to understand complex documents such as SRSs is limited. To help Discovery in this task, we have implemented a conceptual in Knowledge Studio. The implementation is straightforward because Knowledge Studio's model is similar to our *LI* model. Fig. 3 shows how Knowledge Studio works. We have annotated domain-specific documents ("training data"), and the AI generalizes to new documents (the target SRS). More elaborated operations, like k-fold cross-validations, can be performed in this environment as well. To ease annotation, we have created a dictionary containing domain-specific concepts (about the target SRS problem domain).

The machine learning model trained in Knowledge Studio was deployed to Watson Discovery (arrow "1"). The features of the simulator do not change with this trained model. However, Discovery's abilities to uncover the right information are greatly improved.

V. CHATBOT EVALUATION

Different evaluation methods have been applied as recommended by Kitchenham et al. [34]. The evaluation is described in detail in [32]. We provide an abridged description herein.

A. First cycle evaluation

The first cycle's proof of concept was evaluated using *Expert evaluation*. The second author played this role. The first version of the simulator received extensive criticism. The expert raised issues about the definition of almost all intents, the usage of entities (see the technical details in the sections below), and the analysis of the SRS document.

B. Second cycle evaluation

The second cycle was evaluated using expert analysis again. The main issues raised by the expert were: (1) incomplete responses, (2) the appearance of irrelevant text in the responses,

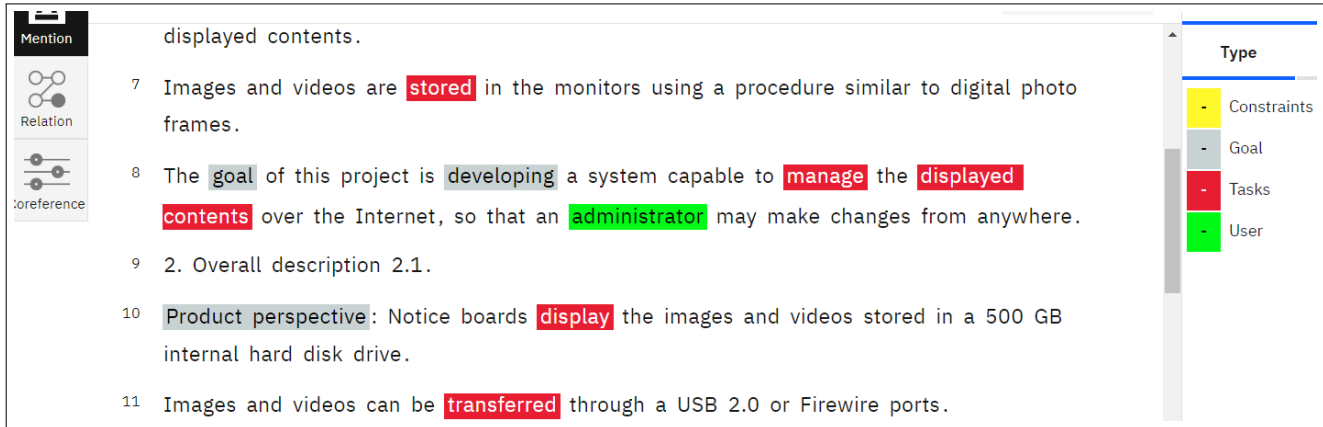


Fig. 3. Knowledge Studio's annotation screen.

(3) issues related to naming conventions, and (4) errors in the definition of entities.

C. Third cycle evaluation

The evaluation of the third cycle relied on *surveys*. The evaluation was conducted by one first-year MSc student with the Universidad Politécnica de Madrid. He has taken a Requirements Engineering course in the semester previous to this evaluation, including (short) interview training. To facilitate the process of evaluation, context-free questions were given to the student. The simulator-student conversation lasted 5 minutes approximately.

The first survey (available at <https://bit.ly/3gcQOVt>) aims to assess the degree to which the student has acquired knowledge about the future software system during the interaction with the simulator. The student was able to answer most (9 out of 10) questions correctly. This result shall be taken with caution due that (1) just one student participated in the evaluation and (2) the Display Management System is rather simple.

The second survey (available at <https://bit.ly/3e7LgcR>) collects the student's impressions regarding the interaction with the simulator (not the knowledge acquired). The responses are given on a 5-point Likert scale. Quite not surprisingly, the simulator has received a very good rating (most of the responses are 4's and 5's).

Both surveys include one open-ended question. The student suggested that the simulator should answer questions more precisely, e.g., in the third cycle's proof of concept, all information related to users is displayed at once. Another recommendation was that the simulator should also provide support for clarification questions.

D. Fourth cycle evaluation

During the evaluation of the fourth cycle, we applied the same approach used in the third cycle. The difference is that three students instead of one participated. Two of them are first-year MSc students with the Universidad Politécnica de

Madrid and the third one is a second-year MSc student at the same university.

The students were able to answer 86 percent of questions correctly. The simulator also received a positive rating (median 4 out of 5). In the open-ended questions, the students highlighted that the simulator performs well but there is still room for some improvements, e.g., the simulator should highlight the entities properly, provide support for making summaries, and understand clarification questions.

E. Fifth cycle evaluation

The fifth cycle used a hybrid approach. First, the correctness and completeness of the information provided by the chatbot were evaluated using empirical study. Second, a survey was used to get the students' feedback. The students who participated in the previous cycle were selected for this cycle.

1) *Hypotheses*: Our research aims to help students to gain interview skills. An interview is successful when all relevant information is correctly acquired. Thus, our hypotheses are:

- $H1_0$: The students cannot retrieve the correct information using the chatbot.
- $H1_1$: The students can retrieve the correct information using the chatbot.

and

- $H2_0$: The students cannot retrieve complete information using the chatbot.
- $H2_1$: The students can retrieve complete information using the chatbot.

2) *Independent Variables*: The independent variable is the chatbot usage. In a full-fledged experiment, this variable would have two different levels or values: using chatbot vs. conducting an actual interview. Due to the prototype character of the chatbot, a comparison with a real interview would be unfair. Likewise, the number of experimental subjects is limited. Therefore, we decided to perform an exploratory quasi-experiment (notice that we do not have the control group) instead of a real experiment. The goal is understanding

TABLE I
QUASI-EXPERIMENT RESULTS

		Identified information	
		Correct items	Wrong items
Subject's SRS	1	17	1
	2	17	2
	3	15	0

whether Hx_0 could be rejected in larger-scale studies, instead of formally testing the hypotheses using statistical methods.

3) *Dependent Variable*: There are two dependent variables:

- The subjects' capability to correctly retrieve information using the chatbot.
- The completeness of the information acquired.

The variables are measured using the target SRS as *gold standard*, e.g., see [35]. The possible values lie in the interval $[0 - 100]$, i.e., they represent the amount (in percentage) of information obtained as compared to the gold standard.

4) *Experimental task*: We asked the students to use the system and create SRS for a system using the simulator. We used the same SRS than in previous evaluations. This represents a threat to validity, but less serious than it seems: the students do not have complete information about the SRS, the interaction with the simulator in the previous cycles had a different purpose, and was time-constrained.

5) *Results*: The results are shown in Table I. We did not perform statistical analysis due to the small sample and the absence of a control group. The total number of items in the gold standard is 24. The *correctness* can be calculated as $100 - \frac{1/17+2/17+0/15}{3} \times 100 = 94\%$. The *completeness* is $\frac{17/24+17/24+15/24}{3} \times 100 = 68\%$. These values are rather satisfactory and reasonably realistic, but they should be taken with caution due to the existing threats to validity and the simplicity of the target SRS. The simulator received a 4.5 out of 5 ratings. All students agreed that the system could be helpful for training novice requirements engineers.

F. Sixth cycle evaluation

The simulator's key characteristics did not change between the fifth and sixth cycles. Therefore, we did not perform an empirical study again. We kept the opinion survey with the only intention of finding out if something went wrong in the sixth cycle's proof of concept. No problems were reported.

VI. THREATS TO VALIDITY

The threats to validity are described according to Runeson et al. [36, p. 71 – 73], whom in turn are based on Yin's seminal work [37]: (1) Construct validity, (2) internal validity, (3) external validity, and (4) reliability.

A. Construct validity

Construct validity refers to the degree of concordance between the researchers' perspective and the objective reality of the phenomenon under study. This threat mainly impacts the instruments used to measure the effectiveness of the interview simulator, such as the surveys mentioned in section V. The

mitigation strategy is provided by the cyclical arrangement of the *Design Science* methodology: The instruments are examined after each cycle, and improvements are introduced before the next cycle.

B. Internal validity

Internal validity refers to incorrectly assign correlations to the category of causal relationships. This threat operates in the achievement of goals 1-3, i.e., pretending that the engineer-simulator conversation is a "natural" conversation when it is not. The mitigation strategy rests on increasing the involvement of RE students and instructors in the evaluations of the proofs of concept. Cycles 1-2 use expert evaluation, cycles 3, 4, and 6 students surveys, and cycle 5 small-size experiments with students. The number and diversity of checks increase our confidence in the simulator's capabilities, but we recognize that the validation has been limited. To fight against this threat, we plan to use the simulator during one semester in a master-level RE course.

C. External validity

External validity aims that the simulator will work in contexts different from the one where it has been initially designed. This threat operates in two directions:

- The interview simulator applies to different problem domains. The modular structure of the simulator (Knowledge studio, Discovery) supports defining requirements problems in different domains. Unfortunately, we have used the simulator in just one problem domain.
- The design theory (goal 4) enables the creation of new simulators, probably using different elicitation techniques. However, creating simulators for different elicitation techniques was out of the scope of this research.

We plan to address these threats opening the simulator code to the community when the one-semester live test finishes with satisfactory results. External replication will be possible since.

D. Reliability

Reliability seeks that our research is described with enough detail to enable replication by external researchers. In this regard, the minute detail of the research is publicly available. The simulator code will be publicly available in March 2021.

VII. CONCLUSION AND FUTURE RESEARCH

The interview simulator has been created using the *Design Science* approach with the primary goal of helping novice requirements engineers. The results have been promising:

- The interview simulator understands context-free questions, retrieving the right information related to RE concepts such as Tasks, Goals, Users, Benefits, Stakeholders, Constraints, and Integration.
- The simulator also makes summaries of the conversation, answer meta-questions, and answer the questions based on the context. Moreover, the simulator also responds randomly with ambiguity and incompleteness.

- The simulator's knowledge base has been defined at 3 levels: (1) *L2*, i.e., classes and relationships, (2) *L1*, RE-related concepts, and (3) *L0*, domain-specific concepts. This knowledge base is easy to implement in Watson using the Knowledge Studio and Assistant services.
- The modular structure of the knowledge base leverages the simulator's ability to maintain conversations in different domains. Code changes will not be likely required.

The future research will focus on applying the interview simulator to different problem domains by utilizing the modular structure of the knowledge base. In addition to different domains, more work will be done to make an interview simulator to act like a real assistant. The interview simulator will be tested in a real RE course in the academic year 2020-2021. Once it proves effective in the classroom, it will be opened to the RE community for free use and improvement.

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